

NORTH FRISIAN SURVEY - DIVE OPERATIONS SAFETY PROTOCOL

Standard Operating Procedures for Underwater Survey and Excavation

Nordmeer Maritime University | DFG Project GR 4471/3

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1. Purpose and Scope

This protocol governs all underwater operations conducted under the North Frisian Medieval Wreck Survey. It applies to every diver, tender, and vessel crew member who participates in survey, inspection, or excavation dives at any project site, including Site NF-7 and any newly identified targets. Compliance with this protocol is mandatory. Deviations may only be authorised by the Field Operations Lead in consultation with the Principal Investigator.

2. Personnel and Qualifications

Role	Minimum qualification	Additional requirements
Field Operations Lead (Dive Master)	CMAS 3 star + commercial diver	Full face, first aid, O2 provider
Principal Investigator (Survey Diver)	CMAS 2 star + dive medical	Annual dive medical certification
Survey Diver	CMAS 2 star + dive medical	Cold water training preferred
Safety Diver	CMAS rescue diver + O2 provider	Must not be primary survey diver
Deck Hand / Tender	Boat licence SBF See	VHF radio certified
Observer / Student	CMAS 1 star minimum	Must pair with CMAS 2 star or above

3. Pre-Dive Checks

The following checks must be completed by the Field Operations Lead before each dive. A written record must be made in the field logbook and signed by both the Field Operations Lead and the Principal Investigator before any team enters the water.

3.1 Equipment Checks

Item	Check	Responsible
Dive cylinders	Pressure above 200 bar; valve functioning	Field Ops Lead
Regulators	Free-flow and inhalation test	Each diver
BCD	Inflation and dump valve test	Each diver
Wetsuit / Drysuit	Seals, zipper, and cuff integrity	Each diver
Dive computer	Battery and display check	Each diver
Underwater lights	Illumination and O-ring check	Each diver
SMB / DSMB	Inflation test; line clear	Each diver
Dive knife	Present and sheathed	Each diver

Emergency O2 kit	Cylinder at 150+ bar; mask present	Safety Diver
First aid kit	Stocked; expiry dates checked	Field Ops Lead
VHF radio	Charged; test call to coast guard	Deck Hand

3.2 Environmental and Operational Checks

Wind: Maximum 4 Beaufort for planned dives at NF-7 depth range. At or above 5 Beaufort, dive is postponed. Forecast, not observed, conditions determine go or no-go at the time of departure.

Tide and current: Dives at NF-7 are planned for the slack water window centred on low water. Dive pairs must enter the water no more than 20 minutes before predicted slack and must surface before current reaches 0.8 knots. Current velocity is estimated from on-site surface observation before descent.

Visibility: Minimum 2.0 metres at the site before descent is authorised. Visibility is assessed by the diver after descent to 5 metres. If visibility is below 2 metres, the dive is aborted.

Vessel position: The survey vessel must be at anchor with a minimum scope of 5:1 and the vessel operator must remain aboard throughout the dive. GPS position of the vessel must be logged at dive commencement.

4. Dive Planning

All dives at Site NF-7 are no-decompression recreational dives within a maximum planned depth of 18 metres and a planned bottom time not exceeding 50 minutes. The Field Operations Lead sets the dive plan for each pair at the pre-dive briefing. A conservative dive profile is expected; diving to the planned limit is not obligatory.

Buddy pairs are assigned by the Field Operations Lead before departure and are not changed without the Field Operations Lead's agreement. The PI dives only with the Field Operations Lead or with a CMAS 2 star or above.

5. Emergency Procedures

5.1 Diver in distress at surface:

Immediately deploy safety diver. Establish voice contact. Administer O2 if required. Call 112 and Coastguard (channel 16) if condition is serious. Maintain diver at surface; do not return to depth. Log all observations.

5.2 Missing diver:

At agreed surface time, if a diver has not surfaced, commence search immediately. Field Operations Lead descends with safety diver. All other divers remain at surface. Coastguard (channel 16) is alerted after 4 minutes if diver cannot be located.

5.3 Equipment failure underwater:

Buddy signals ascent. Slow controlled ascent at no more than 9 metres per minute. Safety stop at 5 metres for minimum 3 minutes. Surface in contact with buddy throughout.

5.4 Vessel breakdown:

Deploy anchor if in danger of drifting. Contact Coastguard via VHF channel 16. Do not cancel dive in progress unless vessel is in immediate danger; divers must complete safety stop before surfacing.

6. Incident Reporting

Any dive incident, including aborts, near-misses, equipment failures, or medical events, must be recorded in the field logbook on the day of occurrence. A formal incident report must be submitted to the PI within 24 hours. Incidents involving injury or a medical event must also be reported to the university safety officer and, where required by permit conditions, to the relevant maritime authority.

7. Permit Conditions

All diving activities at Site NF-7 are conducted under permits issued by the relevant Lower Saxony and Schleswig-Holstein heritage authorities. A copy of all current permits must be carried on the vessel at all times. No excavation may proceed outside the boundaries specified in the current permit, and any unexpected finds must be logged immediately and the heritage authority notified within 24 hours.